SIGNALS AND SYSTEMS

Unit	Content	Hrs./ Week	Marks
1	**Introduction to Signals and Systems** 1.1 Definition of Signals and Systems 1.2 Classification of Signals: Continuous-Time (CT) and Discrete-Time (DT) Signals, Deterministic and Random Signals 1.3 Standard Test Signals: Unit Step, Unit Ramp, Unit Impulse, Sinusoidal, and Exponential Signals 1.4 Signal Operations: Time Scaling, Time Shifting, Time Reversal	5	10
2	**Mathematical Representation of Signals** 2.1 Even and Odd Signals 2.2 Energy and Power of a Signal 2.3 Orthogonality of Signals 2.4 Concept of Convolution and Correlation	6	12
3	**Classification and Properties of Systems** 3.1 Linear and Non-Linear Systems 3.2 Time-Invariant and Time-Variant Systems 3.3 Causal and Non-Causal Systems 3.4 Stable and Unstable Systems 3.5 Impulse Response and Step Response of a System	6	12
4	**Fourier Series and Fourier Transform** 4.1 Introduction to Fourier Series and Its Need 4.2 Trigonometric and Exponential Fourier Series 4.3 Fourier Transform of Continuous-Time and Discrete-Time Signals 4.4 Properties of Fourier Transform 4.5 Application of Fourier Transform in Signal Processing	7	14
5	**Laplace Transform and Z-Transform** 5.1 Introduction to Laplace Transform and Its Applications 5.2 Properties of Laplace Transform 5.3 Inverse Laplace Transform 5.4 Introduction to Z-Transform and Its Properties 5.5 Inverse Z-Transform	7	14
6	**Sampling and Reconstruction of Signals** 6.1 Sampling Theorem and Its Significance 6.2 Nyquist Rate and Aliasing 6.3 Reconstruction of Signals from Samples 6.4 Zero-Order and First-Order Hold Circuits	6	12
7	**System Analysis Using Transfer Functions** 7.1 Definition of Transfer Function 7.2 Concept of Poles and Zeros 7.3 System Stability Analysis 7.4 Block Diagram Representation of Systems	6	12

Unit	Content	Hrs./ Week	Marks
8	**Applications of Signals and Systems in Industrial Automation** 8.1 Signal Processing in Industrial Automation 8.2 Application of Signal Analysis in Control Systems 8.3 Introduction to Digital Signal Processing (DSP) and Its Applications in Industrial Automation	5	14
	TOTAL	**48**	**100**